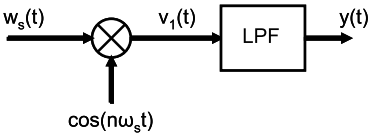
Assignment Chapter 3: Solution

Problem 3-4:



- We perform the analysis in the frequency domain:

$$v_1(t) = w_s(t).\cos(n\omega_s t) \;\;\Rightarrow\;\; V_1(f) = W_s(f) * \left[\frac{1}{2}\delta(f - nf_s) + \frac{1}{2}\delta(f + nf_s)\right]$$

- Consequently:

$$\begin{aligned}
V_1(f) &= d\sum_{k\in\mathbb{Z}} \text{sinc}\,(kd)\, W(f - kf_s) * \left[\frac{1}{2}\delta(f - nf_s) + \frac{1}{2}\delta(f + nf_s)\right] \\
&= \frac{d}{2}\sum_{k\in\mathbb{Z}} \text{sinc}\,(kd)\, W(f - (k+n)f_s) + \frac{d}{2}\sum_{k\in\mathbb{Z}} \text{sinc}\,(kd)\, W(f - (k-n)f_s) \\
&= \frac{d}{2}\text{sinc}\,(-nd)\, W(f) + \underbrace{\frac{d}{2}\sum_{k\in\mathbb{Z},k\neq -n} \text{sinc}\,(kd)\, W(f - (k+n)f_s)}_{(1)} \\
&\quad + \frac{d}{2}\text{sinc}\,(nd)\, W(f) + \underbrace{\frac{d}{2}\sum_{k\in\mathbb{Z},k\neq n} \text{sinc}\,(kd)\, W(f - (k-n)f_s)}_{(2)}
\end{aligned}$$

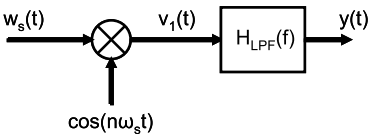
  - The terms in summation (1) are centered at $(k+n)f_s$. Since in this summation, $k \neq -n$, then $k + n \neq 0$ implying that this summation will be eliminated by the LPF.
  - The terms in summation (2) are centered at $(k-n)f_s$. Since in this summation, $k \neq n$, then $k - n \neq 0$ implying that this summation will be eliminated by the LPF.

- Finally, since $\text{sinc}\,(-nd) = \text{sinc}\,(nd)$, then at the output of the LPF:

$$Y(f) = d\text{sinc}\,(nd)\, W(f)$$

  implying that $y(t) = Cw(t)$ with $C = d\text{sinc}\,(nd)$.

Problem 3-5:



- Denote by $H(f) = \tau \text{sinc}(\tau f)$ the Fourier transform of the sampling-pulse shape $h(t) = \text{rect}(t/\tau)$.
- The Fourier transform of the signal at the output of the multiplier is:

$$\begin{aligned}
V_1(f) &= W_s(f) * \left[\frac{1}{2}\delta(f - nf_s) + \frac{1}{2}\delta(f + nf_s)\right] \\
&= \left[\frac{1}{T_s}H(f)\sum_{k\in\mathbb{Z}} W(f - kf_s)\right] * \left[\frac{1}{2}\delta(f - nf_s) + \frac{1}{2}\delta(f + nf_s)\right] \\
&= \frac{1}{2T_s}H(f - nf_s)\sum_{k\in\mathbb{Z}} W(f - (k+n)f_s) + \frac{1}{2T_s}H(f + nf_s)\sum_{k\in\mathbb{Z}} W(f - (k-n)f_s) \\
&= \frac{1}{2T_s}H(f - nf_s)W(f) + \underbrace{\frac{1}{2T_s}H(f - nf_s)\sum_{k\in\mathbb{Z},k\neq -n} W(f - (k+n)f_s)}_{(1)} \\
&\quad + \frac{1}{2T_s}H(f + nf_s)W(f) + \underbrace{\frac{1}{2T_s}H(f + nf_s)\sum_{k\in\mathbb{Z},k\neq n} W(f - (k-n)f_s)}_{(2)}
\end{aligned}$$

- Terms (1) and (2) will not pass though the filter implying that:

$$Y(f) = \frac{1}{2T_s}\left[H(f - nf_s) + H(f + nf_s)\right]W(f)H_{LPF}(f)$$

- For $y(t)$ to be proportional to $w(t)$: $y(t) = Cw(t)$ where C is a constant, then $Y(f) = CW(f)$ <u>in the bandwidth of $w(t)$</u> implying that:

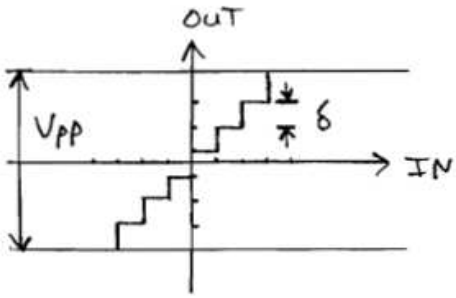
$$\begin{aligned}
&\frac{1}{2T_s}\left[H(f - nf_s) + H(f + nf_s)\right]H_{LPF}(f) = C \\
\Rightarrow\ & H_{LPF}(f) = \frac{2CT_s}{H(f - nf_s) + H(f + nf_s)} = \frac{2C/d}{\text{sinc}(\tau(f - nf_s)) + \text{sinc}(\tau(f + nf_s))}
\end{aligned}$$

- Finally:

$$H_{LPF}(f) = \begin{cases} \frac{2C/d}{\text{sinc}(\tau(f-nf_s))+\text{sinc}(\tau(f+nf_s))}, & |f| < f_c; \\ 0, & \text{elsewhere.} \end{cases}$$

where the cutoff frequency f_c of the filter must satisfy the relation: $B < f_c < f_s - B$ where B stands for the bandwidth of $w(t)$.

Problem 3-7:



- Denote the number of levels by $M = 2^n$ where n is the number of bits in each codeword.
- Denote the peak-to-peak level by V_{pp}, then the step size is given by:

$$\delta = \frac{V_{pp}}{M} = \frac{V_{pp}}{2^n}$$

- On the other hand, the maximum quantization error is equal to $\delta/2$ implying that:

$$\begin{aligned} & \frac{\delta}{2} \leq \frac{P}{100} V_{pp} \\ \Rightarrow \quad & \frac{\delta}{2} \leq \frac{P}{100} 2^n \delta \\ \Rightarrow \quad & 2^n \geq \frac{50}{P} \quad \Rightarrow \quad n \geq \log_2(50/P) \end{aligned}$$

  The term $\log_2(50/P)$ can be written as:

$$\log_2(50/P) = \frac{\ln(50/P)}{\ln(2)}$$

  On the other hand:

$$\log_{10}(50/P) = \frac{\ln(50/P)}{\ln(10)}$$

  Dividing the above equations results in:

$$\frac{\log_2(50/P)}{\log_{10}(50/P)} = \underbrace{\frac{\ln(10)}{\ln(2)}}_{=\log_2(10)} \quad \Rightarrow \quad \log_2(50/P) = \log_2(10) \log_{10}(50/P)$$

- Finally:

$$n \geq \log_2(10) \log_{10}(50/P) = 3.32 \log_{10}(50/P)$$

Problem 3-8:
Part (a):

- The minimum sampling rate is given by:

$$f_s = 2B = 2(100) = 200 \quad \text{samples/sec}$$

Part (b):

- Applying the result obtained in problem 3.7:

$$\begin{aligned} n &\geq 3.22 \log_{10}\left(\frac{50}{P}\right) \\ &= 3.32 \log_{10}\left(\frac{50}{0.1}\right) = 8.96 \end{aligned}$$

  implying that $n = 9$ bits/sample.

Part (c):

- The bit rate of the PCM signal is:

$$\begin{aligned} R &= nf_s \\ &= [9 \quad \text{bits/sample}]\,[200 \text{ samples/sec}] = 1.8 \quad \text{kbits/sec} \end{aligned}$$

Part (d):

- The minimum absolute bandwidth is given by:

$$B = \frac{1}{2}R = 900 \quad \text{Hz}$$

Problem 3-9:

- The signal-to-noise ratio (in decibels) is related to the number of bits/codeword by the relation:

$$\underbrace{\left(\frac{S}{N}\right)_{dB}}_{=30} \simeq 6n \quad \Rightarrow \quad n \simeq 5$$

- The sampling rate is given by:

$$\begin{aligned} R &= nf_s \\ &= [5 \quad \text{bits/sample}]\,[8 \text{ ksamples/sec}] = 40 \quad \text{kbits/sec} \end{aligned}$$

- The time of the stored conversation is:

$$\begin{aligned} T &= \frac{850 \quad \text{Mbytes}}{40 \quad \text{kbits/sec}} \\ &= \frac{850 \times 1024 \times 1024 \times 8 \quad \text{bits}}{40 \times 10^3 \quad \text{bits/sec}} \simeq 178 \times 10^3 \text{ sec} \simeq 49 \text{ hrs}, 31 \text{ min} \end{aligned}$$

Problem 3-19:
Part (a):

- The number of levels L is related to the number of bits l by the relation:

$$L = 2^l = 16 \quad \Rightarrow \quad l = 4 \ \text{ bits/level}$$

Part (b):

- The system transmits one symbol every 0.8 ms implying that the baud rate is given by:

$$D = \frac{N}{T} = \frac{1 \text{ symbol}}{0.8 \times 10^{-3} \text{ sec}} = 1250 \quad \text{baud}$$

Part (c):

- The bit rate is given by:

$$\begin{aligned} R &= lD \\ &= [4 \ \text{ bits/symbol}]\,[1250 \ \text{ symbols/sec}] = 5 \ \text{ kbits/sec} \end{aligned}$$

Problem 3-20:
Part (a):

- The baud rate is given by:

$$D = \frac{R}{l} = \frac{9600 \ \text{ bits/sec}}{4 \ \text{ bits/symbol}} = 2400 \ \text{ baud}$$

  implying that the minimum required bandwidth is:

$$B_{min} = \frac{1}{2} D = 1200 \ \text{ Hz}$$

Part (b):

- For 8-bit encoding:

$$\begin{aligned} D &= \frac{9600}{8} = 1200 \ \text{ baud} \\ \Rightarrow \ B_{min} &= \frac{1}{2} D = 600 \ \text{ Hz} \end{aligned}$$

Problem 3-24:
Part (a):

- For polar signaling the autocorrelation of the data can be calculated from:

$$
\begin{aligned}
k = 0: \quad R(0) &= \sum_{i=1}^{I=2} (a_n a_n)_i P_i \\
&= A^2 \frac{1}{2} + (-A)^2 \frac{1}{2} = A^2 \\
k \neq 0: \quad R(k) &= \sum_{i=1}^{I=4} (a_n a_{n+k})_i P_i \\
&= A^2 \frac{1}{4} + (-A)A\frac{1}{4} + A(-A)\frac{1}{4} + (-A)^2 \frac{1}{4} = 0
\end{aligned}
$$

- The pulse shape is given by $f(t) = \text{rect}(t/\tau)$ implying that: $F(f) = \mathcal{F}[f(t)] = \tau \text{sinc}(\tau f)$.

- Consequently, the PSD is given by:

$$
\begin{aligned}
\mathcal{P}_s(f) &= \frac{A^2}{T_b} |F(f)|^2 = \frac{A^2}{T_b} [\tau \text{sinc}(\tau f)]^2 \\
&= \frac{A^2}{T_b} \left[\frac{T_b}{2} \text{sinc}\left(\frac{T_b}{2} f \right) \right]^2 = \frac{A^2 T_b}{4} \text{sinc}^2 \left(\frac{T_b}{2} f \right)
\end{aligned}
$$

- The null bandwidth is:

$$B_{null} = \frac{2}{T_b} = 2R$$

  implying that the spectral efficiency is:

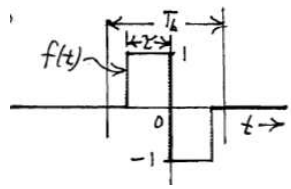
$$\eta = \frac{1}{2} \quad \text{(bit/sec)/Hz}$$

Part (b):

- As with polar signaling, the autocorrelation of the data is given by:

$$R(k) = \begin{cases} A^2, & k = 0; \\ 0, & k \neq 0. \end{cases}$$

- The pulse shape is shown in the figure below:



- Consequently:

$$
\begin{aligned}
f(t) &= \text{rect}\left(\frac{t+\tau/2}{\tau}\right) - \text{rect}\left(\frac{t-\tau/2}{\tau}\right) \\
\Rightarrow \quad F(f) &= \tau\text{sinc}(\tau f)e^{j2\pi f\frac{\tau}{2}} - \tau\text{sinc}(\tau f)e^{-j2\pi f\frac{\tau}{2}} \\
&= 2j\tau\text{sinc}(\tau f)\sin(\pi\tau f)
\end{aligned}
$$

- The PSD is given by:

$$
\begin{aligned}
\mathcal{P}_s(f) &= \frac{A^2}{T_b}|F(f)|^2 = \frac{A^2}{T_b}[4\tau^2\text{sinc}^2(\tau f)\sin^2(\pi\tau f)] \\
&= \frac{A^2}{T_b}\left[4\frac{T_b^2}{16}\text{sinc}^2\left(\frac{T_b f}{4}\right)\sin^2\left(\frac{\pi T_b f}{4}\right)\right] \\
&= \frac{A^2 T_b}{4}\text{sinc}^2\left(\frac{T_b f}{4}\right)\sin^2\left(\frac{\pi T_b f}{4}\right)
\end{aligned}
$$

  - The zero crossings of $\text{sinc}^2\left(\frac{T_b f}{4}\right)$ occur at nonzero multiples of $4/T_b = 4R$.
  - The zero crossings of $\sin^2\left(\frac{\pi T_b f}{4}\right)$ occur at multiples of $4/T_b = 4R$.

- Consequently, the null bandwidth is:

$$
B_{null} = \frac{4}{T_b} = 4R
$$

  implying that the spectral efficiency is:

$$
\eta = \frac{1}{4} \quad \text{(bit/sec)/Hz}
$$

Problem 3-25:

- From lecture 9, the autocorrelation of data with bipolar signaling is given by:

$$R(k) = \begin{cases} \frac{1}{2}A^2, & k = 0; \\ -\frac{1}{4}A^2, & |k| = 1; \\ 0, & |k| > 1. \end{cases}$$

  implying that the PSD is given by:

$$\begin{aligned} \mathcal{P}_s(f) &= \frac{1}{T_b}|F(f)|^2 \left[\frac{A^2}{2} - \frac{A^2}{4}\left(e^{-j2\pi T_b f} + e^{j2\pi T_b f}\right)\right] \\ &= \frac{A^2}{2T_b}|F(f)|^2 \left[1 - \cos(2\pi T_b f)\right] \\ &= \frac{A^2}{T_b}|F(f)|^2 \sin^2(\pi T_b f) \end{aligned}$$

- For bipolar NRZ: $f(t) = \text{rect}(t/T_b)$ implying that $F(f) = T_b\text{sinc}(T_b f)$. Consequently:

$$\mathcal{P}_s(f) = A^2 T_b \text{sinc}^2(T_b f)\sin^2(\pi T_b f)$$

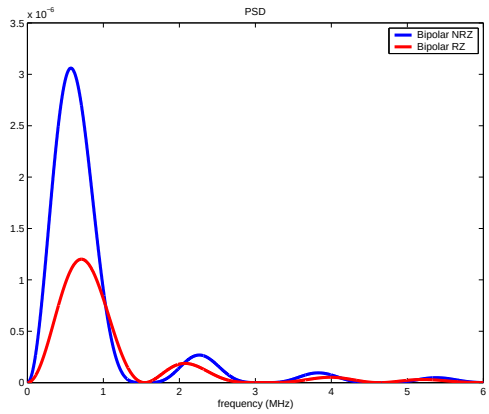
  Note that the null bandwidth is: $B_{null} = 1/T_b = R$.

- For bipolar RZ: $f(t) = \text{rect}(t/\tau)$ implying that $F(f) = \tau\text{sinc}(\tau f) = \frac{T_b}{2}\text{sinc}\left(\frac{T_b}{2}f\right)$. Consequently:

$$\mathcal{P}_s(f) = \frac{A^2 T_b}{4}\text{sinc}^2\left(\frac{T_b}{2}f\right)\sin^2(\pi T_b f)$$

  Note that the null bandwidth is: $B_{null} = 1/T_b = R$.

- The above PSDs are shown in the figure below for $A = 3$ V and $T_b = 1/R = 0.6477\mu$s:



Problem 3-29:

- The encoded sequence $\{e_n\}$ is related to the input sequence $\{d_n\}$ by: $e_n = d_n \oplus e_{n-1}$.
- Depending on the value of the reference bit, the corresponding encoded sequences are as follows:

Input sequence	d_n		0	1	1	0	1	0	0	0	1	0	1
Encoded sequence	e_n	1	1	0	1	1	0	0	0	0	1	1	0
Encoded sequence	e_n	0	0	1	0	0	1	1	1	1	0	0	1

Reference bit

- Note that the encoded sequence is inverted when the reference bit changes its value.

Problem 3-35:
Part (a):

- The number of levels L is related to the number of bits l by the relation:

$$L = 2^l = 8 \quad \Rightarrow \quad l = 3$$

implying that the baud rate is given by:

$$D = \frac{R}{l} = \frac{9600 \ \text{ bits/sec}}{3 \ \text{ bits/symbol}} = 3.2 \ \text{ ksymbols/sec}$$

Part (b):

- The baud rate is related to the absolute bandwidth B by:

$$D = \frac{2B}{1+r} \quad \Rightarrow \quad r = \frac{2B}{D} - 1 = \frac{2 \times 2.4}{3.2} - 1 = 0.5$$

Problem 3-42:

- For $C = 1$ and $\tau = 0$, Nyquist's first method can be written as:

$$h_e(kT_s) = \begin{cases} 1, & k = 0; \\ 0, & k \neq 0. \end{cases}$$

- Consequently:

$$\begin{aligned} h_e(t).\sum_{k=-\infty}^{+\infty} \delta(t - kT_s) &= \sum_{k=-\infty}^{+\infty} h_e(kT_s)\delta(t - kT_s) \\ &= \delta(t) \end{aligned}$$

  where the first equality follows from the multiplication property of the unit impulse and the second equality follows from Nyquist's first method.

- Taking the Fourier transform of both sides of the above equation results in:

$$\begin{aligned} & H_e(f) * \mathcal{F}\left[\sum_{k=-\infty}^{+\infty} \delta(t - kT_s)\right] = 1 \\ \Rightarrow\ & H_e(f) * \frac{1}{T_s}\sum_{k=-\infty}^{+\infty} \delta\left(f - \frac{k}{T_s}\right) = 1 \\ \Rightarrow\ & H_e(f) * \sum_{k=-\infty}^{+\infty} \delta\left(f - \frac{k}{T_s}\right) = T_s \\ \Rightarrow\ & \sum_{k=-\infty}^{+\infty} H_e\left(f - \frac{k}{T_s}\right) = T_s \end{aligned}$$

  Replacing k by $-k$ in the last equation results in:

$$\sum_{k=-\infty}^{+\infty} H_e\left(f + \frac{k}{T_s}\right) = T_s$$

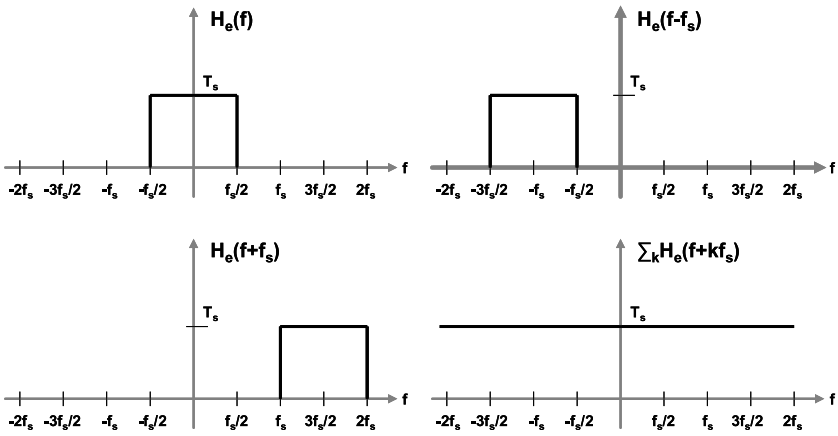
  Note that the Fourier transform of $\sum_k \delta(t - kT_s)$ is $\frac{1}{T_s}\sum_k \delta\left(f - \frac{k}{T_s}\right)$ as shown in lecture 9.

Problem 3-43:
Part (a):

- The relation $f_s = 2f_0$ implies that $T_0 = 2T_s$. Consequently:

$$\begin{aligned} H_e(f) &= \frac{T_0}{2}\text{rect}\left(\frac{1}{2}fT_0\right) \\ &= T_s\text{rect}\left(T_s f\right) \end{aligned}$$

- From the next figure, we observe that $\sum_{k=-\infty}^{+\infty} H_e\left(f + \frac{k}{T_s}\right) = \sum_{k=-\infty}^{+\infty} H_e\left(f + kf_s\right) = T_s$ implying that the filter satisfies Nyquist's criterion for eliminating ISI.



- Note that $h_e(t) = \text{sinc}(t/T_s)$ implying that:

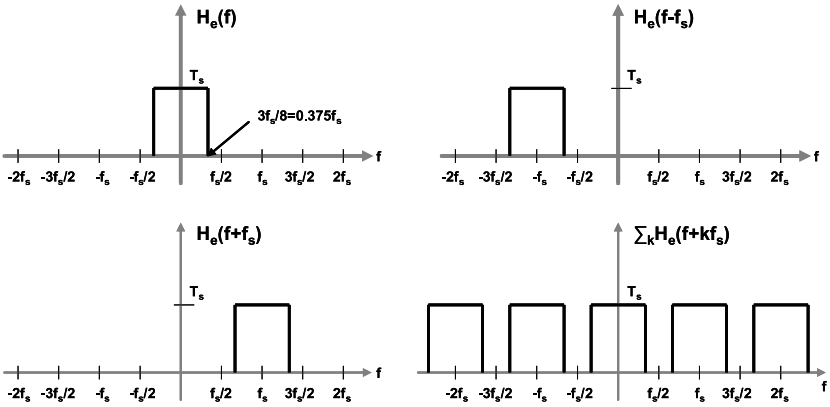
$$h_e(kT_s) = \text{sinc}(k) = \begin{cases} 1, & k = 0; \\ 0, & k \neq 0. \end{cases}$$

Part (b):

- The transfer function of the filter can be written as:

$$\begin{aligned} H_e(f) &= \frac{T_0}{2}\text{rect}\left(\frac{2}{3}fT_0\right) \\ &= T_s\text{rect}\left(\frac{4T_s}{3}f\right) \end{aligned}$$

- From the next figure, we observe that $\sum_{k=-\infty}^{+\infty} H_e\left(f + kf_s\right) \neq T_s$ implying that the filter does not satisfy Nyquist's criterion for eliminating ISI.



- Note that $h_e(t) = \frac{3}{4}\text{sinc}\left(\frac{3}{4T_s}t\right)$ implying that:

$$h_e(kT_s) = \frac{3}{4}\text{sinc}\left(\frac{3}{4}k\right)$$

Note that $h_e(kT_s)$ is not always equal to zero for $k \neq 0$ (for example $h(T_s) = \frac{3}{4}\text{sinc}\left(\frac{3}{4}\right) \neq 0$).

Problem 3-45:
Part (a):

- The baud rate is given by:

$$D = \frac{2B}{1+r} = \frac{2 \times 4 \text{ kHz}}{1+0.5} = 5.33 \quad \text{ksymbols/sec}$$

- Since the PCM signal transmitted over the channel is binary, then the bit rate is given by:

$$R = lD = [1 \quad \text{bit/symbol}][5.33 \quad \text{ksymbols/sec}] = 5.33 \quad \text{kbits/sec}$$

Part (b):

- The bit rate at the output of the quantizer is related to the sampling frequency by:

$$R = nf_s \quad \Rightarrow \quad f_s = \frac{R}{n} = \frac{5.33 \quad \text{kbits/sec}}{4 \quad \text{bits/sample}} = 1.325 \quad \text{ksamples/sec}$$

- Based on the sampling theorem, the sampling frequency must satisfy the following relation:

$$f_s \geq 2B \quad \Rightarrow \quad B \leq \frac{f_s}{2} = 666 \quad \text{Hz}$$

  implying that the maximum permitted bandwidth is $B_{max} = 666$ Hz.

Problem 3-47:
Part (a):

- The baud rate can be calculated from:

$$L = 2^l = 4 \quad \Rightarrow \quad l = 2$$
$$\Rightarrow \quad D = \frac{R}{l} = \frac{2400}{2} = 1200 \quad \text{baud}$$

Part (b):

- The 6-dB bandwidth f_0 is related to the baud rate by:

$$D = 2f_0 \quad \Rightarrow \quad f_0 = \frac{D}{2} = 600 \quad \text{Hz}$$

Part (c):

- The absolute bandwidth B is given by:

$$D = \frac{2B}{1+r} \quad \Rightarrow \quad B = \frac{D(1+r)}{2} = \frac{1200 \times 1.5}{2} = 900 \quad \text{Hz}$$

Problem 3-50:
Part (a):

- The sampling frequency is given by:

$$f_s = 10 f_{Nyq} = 10(2f_{max}) = 20 \times 10 = 200 \ \text{ kHz}$$

- To prevent slope overload noise, the step size δ must satisfy the following relation:

$$\frac{\delta}{T_s} = \delta f_s > A(2\pi f_a)$$
$$\Rightarrow \ \delta > \frac{A2\pi f_a}{f_s} = \frac{0.5 \times 2 \times \pi \times 10}{200} \quad \Rightarrow \quad \delta > 0.157 \ \text{ V}$$

On the other hand, to minimize the granular noise, δ must be kept as small as possible implying that δ must take the following value:

$$\delta = 0.157 \ \text{ V}$$

Part (b):

- The PSD for the granular noise is given by:

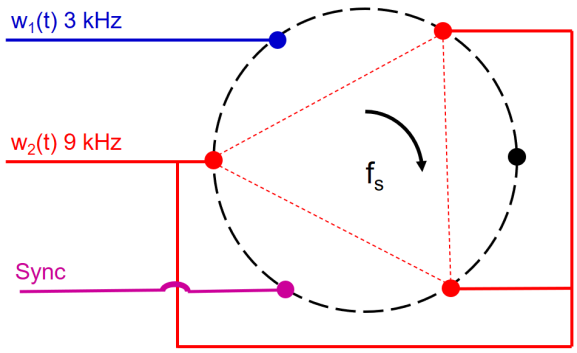
$$\mathcal{P}_n(f) = \frac{\delta^2}{6f_s} = \frac{(0.157)^2}{6 \times 200} = 2.056 \times 10^{-5} \quad \text{V}^2/\text{kHz}$$

Part (c):

- The average signal-to-noise ratio is given by:

$$\left(\frac{S}{N}\right) = \frac{\frac{A^2}{2}}{2B\mathcal{P}_n(f)} = \frac{\frac{0.5^2}{2}}{2 \times 200 \times 2.056 \times 10^{-5}} = 15.19$$
$$\Rightarrow \ \left(\frac{S}{N}\right)_{dB} = 10\log_{10}(15.19) = 11.82 \ \text{ dB}$$

Problem 3-54:
Part (a):



- The minimum sampling frequencies of $w_1(t)$ and $w_2(t)$ are given by:

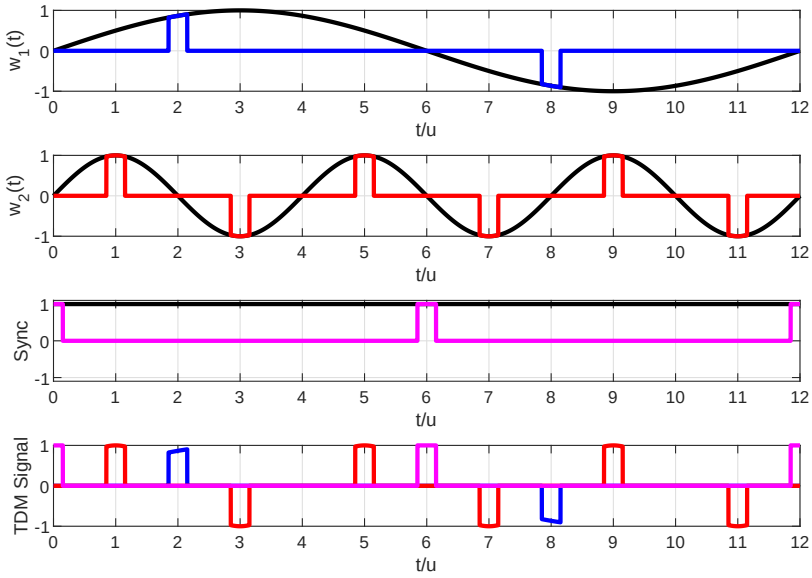
$$f_{s1} = 2 \times 3 = 6 \ \text{ kHz}$$
$$f_{s2} = 2 \times 9 = 18 \ \text{ kHz}$$

- A commutator rotating at $f_c = 6$ kHz can be used to accommodate the above signals. $w_1(t)$ must be connected once to the commutator while $w_2(t)$ must be connected three times to the commutator. In other words, signal $w_1(t)$ is sampled once per rotation while signal $w_2(t)$ is sampled three times per rotation.

- Note that:
  - The 3 symmetrical connections of $w_2(t)$ must be separated by an angle of $2\pi/3$.
  - One synchronization bit is added to the system.
  - The decommutator (at the receiver side) is identical to the commutator (at the transmitter side).

Part (b):

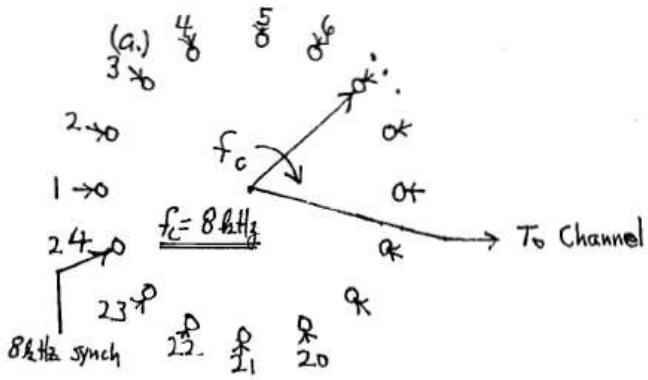
- Typical plots of $w_1(t)$, $w_2(t)$ and the TDM signal are shown in the next figure:



- Note that:
  - The time to complete one rotation of the commutator is $T_s = \frac{1}{f_s} = \frac{1}{6}$ ms.
  - The time to move between two consecutive connections on the commutator is $u = \frac{T_s}{6} = \frac{1}{36}$ ms (this will be considered as the time unit denoted by u).
    - The period (in ms) of $w_1(t)$ is $T_1 = \frac{1}{3} = \frac{12}{36} = 12u$.
    - The period (in ms) of $w_2(t)$ is $T_2 = \frac{1}{9} = \frac{4}{36} = 4u$.

Problem 3-56:
Part (a):

- One commutator rotating at $f_c = 8$ kHz is required. The total number of connections is $23 + 1 = 24$:
  - 1 connection is required for each one of the 23 analog signals resulting in a total of 23 connections.
  - 1 connection is dedicated to the synchronization channel.
- The TDM commutator is shown in the figure below:



- The baud rate of the TDM signal is given by:

$$D = 24 \times 8 = 192 \quad \text{ksamples/sec}$$

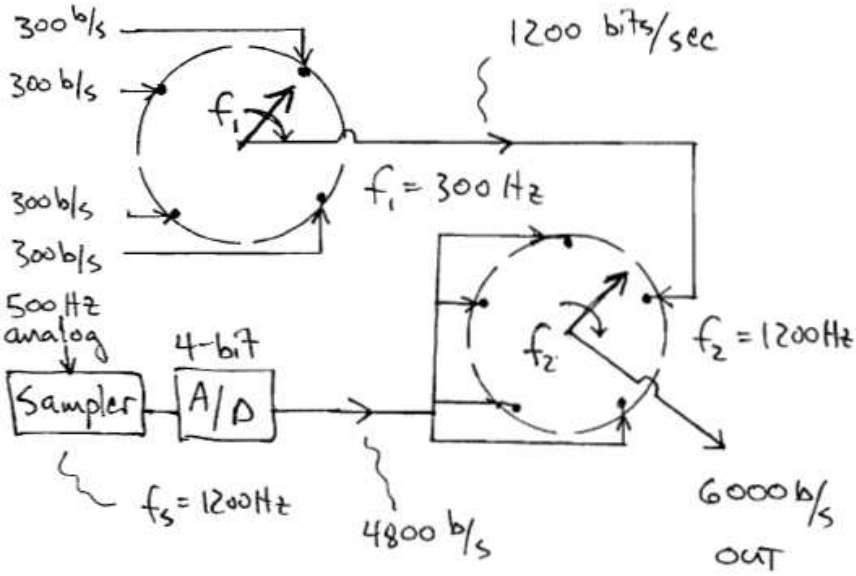
Part (b):

- The absolute bandwidth B can be determined from the following relation:

$$D = \frac{2B}{1+r} \quad \Rightarrow \quad B = \frac{D(1+r)}{2} = \frac{192 \times 1.75}{2} = 168 \ \ \text{kHz}$$

Problem 3-59:

- The TDM system is shown in the figure below:



  - The first commutator rotates at $f_1 = 300$ Hz. Each one of the four 300 bits/s digital sources is connected once resulting in a data rate of $4 \times 300 = 1200$ bits/s at the output of this commutator.
  - The 500 Hz analog source is sampled at 1200 Hz (this sampling frequency corresponds to 1.2 times the Nyquist rate). After a 4-bit A/D, the digitized source will have a data rate of $nf_s = 4 \times 1200 = 4800$ bits/s.
  - The second commutator rotates at $f_2 = 1200$ Hz. The total number of connections on this commutator is $1 + 4 = 5$ resulting in an output bit rate of $5 \times 1200 = 6000$ bits/s.

Problem 3-60:

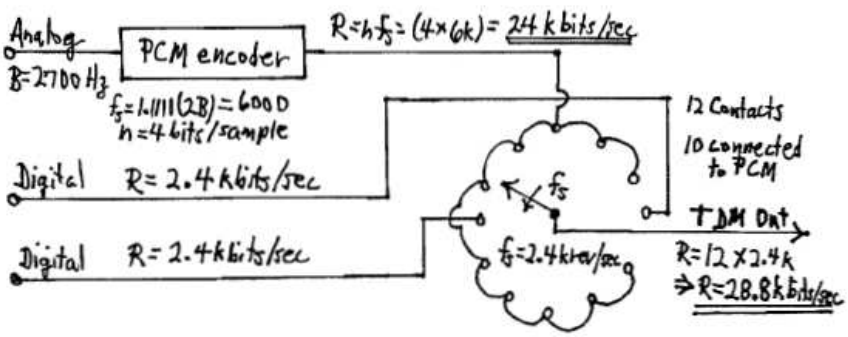
- The analog source is sampled at the rate f_s such that:

  $$f_s = 1.11111 \times f_{Nyq} = 1.11111(2B) = 1.11111 \times 2 \times 2.7 = 6 \ \text{ ksamples/sec}$$

  implying that the data rate of the PCM signal corresponding to this source is given by:

  $$R = nf_s = 4 \times 6 = 24 \ \text{ kbits/sec}$$

- The TDM system accommodating the three sources is shown in the figure below:



- Note that one commutator rotating at $f_c = 2.4$ kHz is required. The total number of connections on the commutator is $2 + 10 = 12$
  - 1 connection is required for each one of the 2 digital sources resulting in a total of 2 connections.
  - 10 connections are required to the digitized analog source having a data rate of 24 kbits/sec.

- Consequently, the bit rate of the TDM signal is given by:

  $$R = 12 \times 2.4 = 28.8 \ \text{ kbits/sec}$$